

Bocconi

BREAK-EVEN POINT ANALYSIS



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Managers at “Roundedges”, a producer of low cost tablets, is planning to expand its production facility due to an explosion in demand from young consumers in Europe. They are currently evaluating the options described below

	Plant A	Plant B
Total investment (expected life = 25 years)	€ 70,000,000	€ 90,000,000
Maximum capacity	2,500,000	3,500,000
Maintenance	€ 7,300,000	€ 9,500,000
Overheads	€ 4,100,000	€ 4,700,000
Indirect labour	€ 1,900,000	€ 2,700,000
Direct labour cost	€ 23.00	€ 22.00
Materials	€ 38.00	€ 35.00
Energy	€ 11.70	€ 11.50



In line with the market forecasts, they are expecting to sell 1,300,000 tablets at a price of €95 each.

- a) Which plant should they invest in? Why?
- b) Suppose the estimates of future demand are now unreliable. Managers still want to sell the tablets at a price of €95. Which plant should they decide to invest in now?
- c) Assume that managers want to reach a target net income of 30,000,000. The tablets are sold at a unitary price of € 95. The firm has a loan of € 20,000,000 at 5% interest rate, and taxes unrelated to income equal to 9,000,000. Which plant should they invest in now?



Managers at “Smarter phone”, a producer of low cost smartphones, is planning to expand its production facility due to an explosion in demand from young consumers in Europe. They are currently evaluating the options described below.

	Plant A	Plant B
Total investment (expected life = 20 years)	€ 75,000,000	€ 94,000,000
Maximum capacity	2,500,000	3,500,000
Maintenance	€ 7,800,000	€ 9,500,000
Overheads	€ 4,100,000	€ 4,700,000
Indirect labour	€ 1,900,000	€ 2,700,000
Direct labour cost	€ 23.00	€ 22.00
Materials	€ 41.00	€ 38.00
Energy	€ 11.70	€ 11.50



In line with the market forecasts, they are expecting to sell 1,800,000 tablets at a price of €105 each.

- a) Compare the two options and advise the management. Which plant should they invest in? Why?
- b) Suppose the estimates of future demand are now unreliable. Managers still want to sell the tablets at a price of €105. Which plant should they decide to invest in now?
- c) Assume that managers want to reach a target net income of 50,000,000. The smartphones are sold at a unitary price of € 105. The firm has loans of € 20,000,000 at 10% interest rate, and taxes unrelated to income equal to 13,000,000. Which plant should they invest in now?

